Hosting the Lung Histopathology Triage MVP

Deployment assessment - verified against the repository at C:\Users\Abdurrahman\Documents\DeepSeekHarness-Projects\LungHistopathology-Triage

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1. What is actually blocking you

There are two backends in the repo, and the one wired up for deployment is broken.

	api.py (root)	backend/main.py
Model class	LC25000CNN (Ghost-141, from pretrained_cnn.py)	SimpleCNN (3 conv layers, defined inline)
Label order	adenocarcinoma, benign, squamous_cell_carcinoma	0: Benign, 1: Adenocarcinoma, 2: Squamous
Verified	PASS - smoke test, live on :8000	FAIL - cannot load either checkpoint

Proven failure: loading model.pth into SimpleCNN raises "Unexpected key(s) in state_dict: backbone.conv1.weight, backbone.layer1.0..." - that checkpoint is a ResNet50 (model.py), not SimpleCNN. So railway.toml (cd backend && python main.py), backend/Dockerfile and DEPLOYMENT.md all point at an app that can never work with your weights.

- **Silent failure mode:** on Windows it crashes printing an emoji (cp1252); on a Linux container it boots successfully with no model loaded, so it serves nonsense instead of erroring.
- **No weights in the repo:** both .pth files were untracked in git, so any deploy from git arrives with zero model files.
- **Minor:** root requirements.txt is missing fastapi and uvicorn (present only in backend/requirements.txt); lung-dashboard/.env points at http://localhost:8000 and must become the live API URL.

2. Why hosting is a real decision here

torch occupies **490 MB on disk** in the virtual environment. That single fact eliminates the obvious free options.

Render (free) - 0.1 CPU / 512 MB RAM. There is a documented case of exactly this stack (FastAPI + a local model + React) exceeding 512 MB on import. Free services also sleep, so a first visitor waits roughly a minute.

Hugging Face Spaces (CPU Basic) - 2 vCPU / 16 GB RAM / 50 GB disk. The only free option that comfortably fits torch. **Unresolved uncertainty:** HF documentation states Docker and Gradio Spaces now require a paid plan while only Static Spaces are free, yet other HF pages still list CPU Basic as free. Verify directly before committing to this path.

3. Recommended split

Frontend to Vercel (free). lung-dashboard/ is a plain Vite app; npm run build produces a static dist/. Nothing about it needs a server. Set VITE_API_BASE to the backend URL as a build-time environment variable.

Backend to Hugging Face Spaces (Docker) if still free, otherwise Railway hobby tier (about $5/month). Keep torch. The Grad-CAM heatmap is the demo's real impact, and it needs gradients - the one thing lost by optimising torch away.

Alternative worth knowing: export LC25000CNN to ONNX and swap torch for onnxruntime (about 50 MB instead of 490 MB). That fits Render's free 512 MB at zero cost, but it removes Grad-CAM, because ONNX Runtime gives no backward pass. Take this path only if the heatmap is not essential to the pitch.

4. Two things to do first, on any host

- **Fix the deploy path to target api.py** - either rewrite backend/main.py to import LC25000CNN from pretrained_cnn.py, or repoint the Dockerfile entrypoint at `uvicorn api:app`. Until this is done, nothing deployed will serve correct predictions.
- **Decide how the model travels.** LungDisease_Classifier.pth is only 19 MB - small enough to commit to git directly (GitHub warns above 50 MB, blocks above 100 MB). The 94 MB model.pth should stay out. This means reversing half of the earlier `git rm --cached`: re-track the 19 MB checkpoint, leave the 94 MB one untracked.

5. Open questions that change the answer

- What is the real monthly ceiling - does about $5/month work, or must it be $0?
- Does the hosted demo need the Grad-CAM heatmap, or is predicted class plus confidence enough?
- Who is the audience - investors, clinicians, or a course submission? That decides whether 60-second cold starts are acceptable.

Note: this report also corrects two earlier errors. backend/ does exist and DEPLOYMENT.md is structurally accurate; it was WORKSPACE_SUMMARY.md (line 24) that was wrong - the same file that wrongly claimed vault/spec.md was missing.